

magnitude faster

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Sinkhorn Algorithm

1. $\mathbf{K} \leftarrow \exp\{-\mathbf{C}/\varepsilon\}$

2. While not converged:

$$\mathbf{u} \leftarrow \mathbf{a} \oslash \mathbf{K} \mathbf{v}$$

$$\mathbf{v} \leftarrow \mathbf{b} \oslash \mathbf{K}^T \mathbf{u}$$

Complexity of Sinkhorn

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- Each iteration = two matrix-vector multiplications + vector division: $O(n^2)$
- Typically converges in $O(1/\varepsilon^2)$ iterations (empirical).
- Compared with LP solvers: **Sinkhorn often orders of magnitude faster.**

Empirical Comparison EOT vs OT

